

# NETWORKS AND COMPLEXITY

## Solution 15-6

*This is an example solution from the forthcoming book Networks and Complexity.  
Find more exercises at <https://github.com/NC-Book/NCB>*

### Ex 15.6: aSIS invasion threshold [Lvl. 3]

The aSIS model is described by the following differential equation system:

$$\begin{aligned}\dot{[I]} &= p[SI] - r[I] \\ \dot{[SS]} &= r[SI] - p[SSI] + w[SI] \\ \dot{[II]} &= p[SI] + 2p[ISI] - 2r[II]\end{aligned}$$

along with the two conservation laws  $[S] + [I] = 1$  and  $[SS] + [II] + [SI] = z/2$  and the closure approximations

$$[SSI] = \frac{2[SS][SI]}{[S]} \quad [ISI] = \frac{[SI]^2}{2[S]}.$$

- a) Compute the Jacobian matrix  $\mathbf{J}$  for an unspecified steady state.

#### Solution

To make our life a bit easier we refrain from substituting everything right away. Since we have chosen  $[I]$ ,  $[SS]$ , and  $[II]$  as our variables, we will treat  $[S]$  and  $[SI]$  as auxiliary variables. This means they are just shorthand notations for the expressions

$$[S] = 1 - [I] \tag{1}$$

$$[SI] = z/2 - [SS] - [II] \tag{2}$$

When we compute the derivatives for the Jacobi matrix, we will have to differentiate these functions accordingly. Their derivatives with respect to the variables are

$$\frac{\partial}{\partial [I]}[S] = -1 \quad \frac{\partial}{\partial [SS]}[S] = 0 \quad \frac{\partial}{\partial [II]}[S] = 0 \tag{3}$$

$$\frac{\partial}{\partial [I]}[SI] = 0 \quad \frac{\partial}{\partial [SS]}[SI] = -1 \quad \frac{\partial}{\partial [II]}[SI] = -1 \tag{4}$$

Likewise we will need the derivatives of the closure approximations

$$[SSI] = \frac{2[SS][SI]}{[S]} \quad [ISI] = \frac{[SI]^2}{2[S]}. \tag{5}$$

We compute

$$\frac{\partial}{\partial [I]}[SSI] = -\frac{2[SS][SI]}{[S]^2} \frac{\partial [S]}{\partial [I]} = \frac{2[SS][SI]}{[S]^2} \tag{6}$$

$$\frac{\partial}{\partial [SS]}[SSI] = \frac{2[SI]}{[S]} - \frac{2[SS]}{[S]} = \frac{2([SI] - [SS])}{[S]} \tag{7}$$

$$\frac{\partial}{\partial [II]}[SSI] = -\frac{2[SS]}{[S]} \tag{8}$$

$$\frac{\partial}{\partial[I]}[ISI] = \frac{[SI]^2}{2[S]^2} \quad (9)$$

$$\frac{\partial}{\partial[SS]}[ISI] = -\frac{[SI]}{[S]} \quad (10)$$

$$\frac{\partial}{\partial[II]}[ISI] = -\frac{[SI]}{[S]} \quad (11)$$

We can now compute the nine derivatives for the Jacobian

$$\frac{\partial[\dot{I}]}{\partial[I]} = -r \quad (12)$$

$$\frac{\partial[\dot{I}]}{\partial[SS]} = -p \quad (13)$$

$$\frac{\partial[\dot{I}]}{\partial[II]} = -p \quad (14)$$

$$\frac{\partial[\dot{S}S]}{\partial[I]} = -\frac{2p[SS][SI]}{[S]^2} \quad (15)$$

$$\frac{\partial[\dot{S}S]}{\partial[SS]} = -r - \frac{2p([SI] - [SS])}{[S]} - w \quad (16)$$

$$\frac{\partial[\dot{S}S]}{\partial[II]} = -r + \frac{2p[SS]}{[S]} - w \quad (17)$$

$$\frac{\partial[\dot{I}I]}{\partial[I]} = \frac{p[SI]^2}{[S]^2} \quad (18)$$

$$\frac{\partial[\dot{I}I]}{\partial[SS]} = -p - \frac{2p[SI]}{[S]} \quad (19)$$

$$\frac{\partial[\dot{I}I]}{\partial[II]} = -p - \frac{2p[SI]}{[S]} - 2r \quad (20)$$

After this nice exercise in differentiation we have our Jacobian

$$\mathbf{J} = \begin{pmatrix} -r & -p & -p \\ -\frac{2p[SS][SI]}{[S]^2} & -r - \frac{2p([SI] - [SS])}{[S]} - w & -r + \frac{2p[SS]}{[S]} - w \\ \frac{p[SI]^2}{[S]^2} & -p - \frac{2p[SI]}{[S]} & -p - \frac{2p[SI]}{[S]} - 2r \end{pmatrix}. \quad (21)$$

- b) Find a condition for the *invasion threshold*, the critical value of  $p$  beyond which the disease can invade a disease-free system. [Use  $|\mathbf{J}| = 0$ ].

Solution

**Condition for the invasion threshold.** In the healthy state we know

$$[I] = 0 \quad [S] = 1 \quad (22)$$

$$[SI] = 0 \quad [II] = 0 \quad [SS] = z/2 \quad (23)$$

Substituting this steady state into our Jacobian simplifies it considerably and we are left with

$$\mathbf{J} = \begin{pmatrix} -r & -p & -p \\ 0 & -r + pz - w & -r + pz - w \\ 0 & -p & -p - 2r \end{pmatrix}. \quad (24)$$

Because the diagonal element  $J_{11} = -r$  is alone in its column, it must be also an eigenvalue,  $\lambda_1 = -r$ . So one way to make a bifurcation occur is to set  $r = 0$ . This makes sense, if the nodes no longer recover the system changes fundamentally. So it is good that this turns out to be a bifurcation, but it is not the one we are after. To check for other ways in which an eigenvalue can become zero, we consider the  $2 \times 2$  block that is left when we remove the first row and column

$$\mathbf{J}_{\overline{11}} = \begin{pmatrix} -r + pz - w & -r + pz - w \\ -p & -p - 2r \end{pmatrix} \quad (25)$$

As we compute the determinant we want to exploit that one line has two identical elements

$$|\mathbf{J}_{\overline{11}}| = (-r + pz - w)(-p - 2r) - (-r + pz - w)(-p) \quad (26)$$

$$= (-r + pz - w)(-p - 2r + p) \quad (27)$$

$$= (-r + pz - w)(2r) \quad (28)$$

Since we are looking for solutions where the determinant becomes zero, multiplying this out would be a step in the wrong direction. We already know that a bifurcation happens at  $r = 0$  so we are no longer interested in this. Once we exclude  $r = 0$ , it is safe to divide by  $2r$  and we are left with the condition

$$-r + pz - w = 0 \quad (29)$$

This simplified nicely. It is also good to see that  $p$  and  $z$  only appear as a product. You'd expect that the number of links and the transmission rate across those links somehow multiply. It is also interesting to see that rewiring acts exactly as recovery. This was unexpected but is a nice insight into the system.

**Critical value of  $p$ .** This is simple now. We solve our condition for  $p$ , which yields

$$p = \frac{r + w}{z} \quad (30)$$

and of course for  $w = 0$  we recover

$$p = \frac{r}{z} \quad (31)$$

which is consistent with the previous result.

As a side note, we find that our social distancing strategy is quite effective in preventing the invasion, and we are not even really distancing. As a healthy person you are not isolating, you are just meeting other people.